

Sebeni morphotactic post-training on MULTI13: SFT, DPO, APO and GRPO arms with SmolLM2-135M

Sebastien Diarra

Submitted by Sebastien Diarra

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Official implementation · Research paper · Natural language processing

Abstract

Three post-training arms of Sebeni, a morphotactic generation toolkit for extremely low-resource Manding languages, on the MULTI13 scope (13 languages). Each arm distills the grammar and dictionary, fine-tunes SmolLM2-135M with 4-bit LoRA for 500 steps on the packaged 300-sample dataset with a 10% held-out split, and scores the held-out test set (phi, MER, MCS, UWEC). Code: github.com/mlsftwrs/sebeni at commit 8e66775, with K-Veritas capture added to the CLI.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], [3], [4], [5], which remain authoritative. This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00001v2>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	SFT arm	2026-10-05	4	4	Signed by the K-Veritas server
r2 [3]	DPO arm	2026-10-05	4	4	Signed by the K-Veritas server
r3 [4]	APO arm	2026-10-05	4	4	Signed by the K-Veritas server
r4 [5]	GRPO arm	2026-10-10	4	4	Signed by the K-Veritas server

2 Results

r1: SFT arm

Run	Metric	Final value	Step
3	loss	1.30145	500
3	eval_loss	1.32598	500
3	eval_runtime	22.1618	500
3	eval_samples_per_second	16.244	500
3	eval_steps_per_second	2.031	500
3	eval_entropy	1.36176	500
3	eval_num_tokens	1.52793e+07	500
3	eval_mean_token_accuracy	0.768744	500
3	epoch	2.4642	500

Run	Metric	Final value	Step
4	phi_bam	0.823658	0
4	mer_bam	0	0
4	mcs_bam	0	0
4	uwec_bam	1	0
4	phi_bmq	0.752429	0
4	mer_bmq	3	0
4	mcs_bmq	0.833333	0
4	uwec_bmq	0.666667	0
4	phi_boz	0.916823	0
4	mer_boz	0	0
4	mcs_boz	0	0
4	uwec_boz	1	0
4	phi_dtm	0.809387	0
4	mer_dtm	0.5	0
4	mcs_dtm	0.333333	0
4	uwec_dtm	0.666667	0
4	phi_ful	0.980054	0
4	mer_ful	3	0
4	mcs_ful	0.75	0
4	uwec_ful	0.625	0
4	phi_mey	0.768726	0
4	mer_mey	0	0
4	mcs_mey	0	0
4	uwec_mey	0.5	0
4	phi_mku	0.908601	0
4	mer_mku	12	0
4	mcs_mku	0.923077	0
4	uwec_mku	0.538462	0
4	phi_kao	0.970238	0
4	mer_kao	25.3333	0
4	mcs_kao	0.974359	0
4	uwec_kao	1	0
4	phi_myk	0.7953	0
4	mer_myk	10	0
4	mcs_myk	0.909091	0
4	uwec_myk	0.5	0
4	phi_ses	0.836952	0
4	mer_ses	0	0
4	mcs_ses	0	0
4	uwec_ses	1	0
4	phi_spp	0.853018	0
4	mer_spp	6.66667	0
4	mcs_spp	0.909091	0
4	uwec_spp	0.545455	0
4	phi_snk	0.781673	0
4	mer_snk	0.333333	0
4	mcs_snk	0.333333	0
4	uwec_snk	0.666667	0
4	phi_taq	0.84844	0
4	mer_taq	0.5	0

Run	Metric	Final value	Step
4	mcs_taq	0.5	0
4	uwec_taq	1	0
4	phi	0.849638	0
4	mer	5.16667	0
4	mcs	0.865169	0
4	uwec	0.769663	0

Run	Inline claim	Value	Phase
4	phi	0.849638	eval
4	mer	5.16667	eval
4	mcs	0.865169	eval
4	uwec	0.769663	eval

r2: DPO arm

Run	Metric	Final value	Step
3	loss	0.544094	500
3	rewards_chosen	0.191217	500
3	rewards_rejected	-0.234148	500
3	rewards_accuracies	0.6875	500
3	rewards_margins	0.425366	500
3	eval_rewards_chosen	-0.00811578	500
3	eval_rewards_rejected	-0.419341	500
3	eval_rewards_accuracies	0.719444	500
3	eval_rewards_margins	0.411225	500
3	eval_loss	0.54551	500
3	eval_runtime	77.3164	500
3	eval_samples_per_second	4.656	500
3	eval_steps_per_second	0.582	500
3	eval_entropy	0.887941	500
3	eval_num_tokens	3.0435e+07	500
3	eval_logits_chosen	10.1072	500
3	eval_logits_rejected	10.1393	500
3	eval_mean_token_accuracy	0.850163	500
3	eval_logps_chosen	-1112.08	500
3	eval_logps_rejected	-1114.43	500
3	epoch	2.4642	500
3	phi	0.847792	500
3	phi_bam	0.947662	500
3	phi_bmq	0.788389	500
3	phi_mku	0.874554	500
3	phi_myk	0.858518	500
3	phi_ses	0.673515	500
3	phi_dtm	0.875799	500
3	phi_snk	0.839542	500
3	phi_boz	0.880553	500
3	phi_ful	0.872129	500
3	phi_spp	0.999123	500
3	phi_taq	0.715931	500
4	phi_bam	0.823658	0
4	mer_bam	0	0

Run	Metric	Final value	Step
4	mcs_bam	0	0
4	uwec_bam	1	0
4	phi_bmq	0.752429	0
4	mer_bmq	3	0
4	mcs_bmq	0.833333	0
4	uwec_bmq	0.666667	0
4	phi_boz	0.916823	0
4	mer_boz	0	0
4	mcs_boz	0	0
4	uwec_boz	1	0
4	phi_dtm	0.809387	0
4	mer_dtm	0.5	0
4	mcs_dtm	0.333333	0
4	uwec_dtm	0.666667	0
4	phi_ful	0.980054	0
4	mer_ful	3	0
4	mcs_ful	0.75	0
4	uwec_ful	0.625	0
4	phi_mey	0.768726	0
4	mer_mey	0	0
4	mcs_mey	0	0
4	uwec_mey	0.5	0
4	phi_mku	0.908601	0
4	mer_mku	12	0
4	mcs_mku	0.923077	0
4	uwec_mku	0.538462	0
4	phi_kao	0.970238	0
4	mer_kao	25.3333	0
4	mcs_kao	0.974359	0
4	uwec_kao	1	0
4	phi_myk	0.7953	0
4	mer_myk	10	0
4	mcs_myk	0.909091	0
4	uwec_myk	0.5	0
4	phi_ses	0.836952	0
4	mer_ses	0	0
4	mcs_ses	0	0
4	uwec_ses	1	0
4	phi_spp	0.853018	0
4	mer_spp	6.66667	0
4	mcs_spp	0.909091	0
4	uwec_spp	0.545455	0
4	phi_snk	0.781673	0
4	mer_snk	0.333333	0
4	mcs_snk	0.333333	0
4	uwec_snk	0.666667	0
4	phi_taq	0.84844	0
4	mer_taq	0.5	0
4	mcs_taq	0.5	0
4	uwec_taq	1	0

Run	Metric	Final value	Step
4	phi	0.849638	0
4	mer	5.16667	0
4	mcs	0.865169	0
4	uwec	0.769663	0

Run	Inline claim	Value	Phase
4	phi	0.849638	eval
4	mer	5.16667	eval
4	mcs	0.865169	eval
4	uwec	0.769663	eval

r3: APO arm

Run	Metric	Final value	Step
3	loss	0.899064	500
3	rewards_chosen	0.491311	500
3	rewards_rejected	0.0567837	500
3	rewards_accuracies	0.8125	500
3	rewards_margins	0.434527	500
3	eval_rewards_chosen	0.295494	500
3	eval_rewards_rejected	-0.0857438	500
3	eval_rewards_accuracies	0.669444	500
3	eval_rewards_margins	0.381238	500
3	eval_loss	0.911244	500
3	eval_runtime	77.3757	500
3	eval_samples_per_second	4.653	500
3	eval_steps_per_second	0.582	500
3	eval_entropy	0.887046	500
3	eval_num_tokens	3.0435e+07	500
3	eval_logits_chosen	10.1392	500
3	eval_logits_rejected	10.1657	500
3	eval_mean_token_accuracy	0.85068	500
3	eval_logps_chosen	-1109.39	500
3	eval_logps_rejected	-1111.44	500
3	epoch	2.4642	500
3	phi	0.847792	500
3	phi_bam	0.947662	500
3	phi_bmqr	0.788389	500
3	phi_mku	0.874554	500
3	phi_myk	0.858518	500
3	phi_ses	0.673515	500
3	phi_dtm	0.875799	500
3	phi_snk	0.839542	500
3	phi_boz	0.880553	500
3	phi_ful	0.872129	500
3	phi_spp	0.999123	500
3	phi_taq	0.715931	500
4	phi_bam	0.823658	0
4	mer_bam	0	0
4	mcs_bam	0	0
4	uwec_bam	1	0

Run	Metric	Final value	Step
4	phi_bmq	0.752429	0
4	mer_bmq	3	0
4	mcs_bmq	0.833333	0
4	uwec_bmq	0.666667	0
4	phi_boz	0.916823	0
4	mer_boz	0	0
4	mcs_boz	0	0
4	uwec_boz	1	0
4	phi_dtm	0.809387	0
4	mer_dtm	0.5	0
4	mcs_dtm	0.333333	0
4	uwec_dtm	0.666667	0
4	phi_ful	0.980054	0
4	mer_ful	3	0
4	mcs_ful	0.75	0
4	uwec_ful	0.625	0
4	phi_mey	0.768726	0
4	mer_mey	0	0
4	mcs_mey	0	0
4	uwec_mey	0.5	0
4	phi_mku	0.908601	0
4	mer_mku	12	0
4	mcs_mku	0.923077	0
4	uwec_mku	0.538462	0
4	phi_kao	0.970238	0
4	mer_kao	25.3333	0
4	mcs_kao	0.974359	0
4	uwec_kao	1	0
4	phi_myk	0.7953	0
4	mer_myk	10	0
4	mcs_myk	0.909091	0
4	uwec_myk	0.5	0
4	phi_ses	0.836952	0
4	mer_ses	0	0
4	mcs_ses	0	0
4	uwec_ses	1	0
4	phi_spp	0.853018	0
4	mer_spp	6.66667	0
4	mcs_spp	0.909091	0
4	uwec_spp	0.545455	0
4	phi_snk	0.781673	0
4	mer_snk	0.333333	0
4	mcs_snk	0.333333	0
4	uwec_snk	0.666667	0
4	phi_taq	0.84844	0
4	mer_taq	0.5	0
4	mcs_taq	0.5	0
4	uwec_taq	1	0
4	phi	0.849638	0
4	mer	5.16667	0

Run	Metric	Final value	Step
4	mcs	0.865169	0
4	uwec	0.769663	0

Run	Inline claim	Value	Phase
4	phi	0.849638	eval
4	mer	5.16667	eval
4	mcs	0.865169	eval
4	uwec	0.769663	eval

r4: GRPO arm

Run	Metric	Final value	Step
3	loss	6.85402e-05	500
3	rewards_reward_format_mean	0	500
3	rewards_reward_format_std	0	500
3	rewards_reward_morph_mean	0	500
3	rewards_reward_morph_std	0	500
3	rewards_reward_rule_mean	0	500
3	rewards_reward_rule_std	0	500
3	rewards_reward_lang_mean	0	500
3	rewards_reward_lang_std	0	500
3	reward	0	500
3	frac_reward_zero_std	1	500
3	reward_format	0	500
3	reward_morph	0	500
3	reward_rule	0	500
3	reward_lang	0	500
3	reward_total	2.35746	500
3	eval_rewards_reward_format_mean	0.000347222	500
3	eval_rewards_reward_format_std	0.000982093	500
3	eval_rewards_reward_morph_mean	0.000287138	500
3	eval_rewards_reward_morph_std	0.000812148	500
3	eval_rewards_reward_rule_mean	0.00100277	500
3	eval_rewards_reward_rule_std	0.00283625	500
3	eval_rewards_reward_lang_mean	0	500
3	eval_rewards_reward_lang_std	0	500
3	eval_reward	0.00163713	500
3	eval_reward_std	0.00463049	500
3	eval_frac_reward_zero_std	0.961111	500
3	eval_loss	0.00190403	500
3	eval_runtime	11047.6	500
3	eval_samples_per_second	0.033	500
3	eval_steps_per_second	0.004	500
3	eval_num_tokens	1.23798e+07	500
3	eval_completions_mean_length	994.831	500
3	eval_completions_min_length	822.778	500
3	eval_completions_max_length	1024	500
3	eval_completions_clipped_ratio	0.936111	500
3	eval_completions_mean_terminated_length	235.125	500
3	eval_completions_min_terminated_length	219.756	500
3	eval_completions_max_terminated_length	249.628	500

Run	Metric	Final value	Step
3	eval_kl	0.000742825	500
3	eval_entropy	0.562962	500
3	eval_clip_ratio_low_mean	0	500
3	eval_clip_ratio_high_mean	0	500
3	eval_clip_ratio_region_mean	0	500
3	eval_clip_ratio_low_min	0	500
3	eval_clip_ratio_high_max	0	500
3	epoch	0.617284	500
3	phi	0.847792	500
3	phi_bam	0.947662	500
3	phi_bmq	0.788389	500
3	phi_mku	0.874554	500
3	phi_myk	0.858518	500
3	phi_ses	0.673515	500
3	phi_dtm	0.875799	500
3	phi_snk	0.839542	500
3	phi_boz	0.880553	500
3	phi_ful	0.872129	500
3	phi_spp	0.999123	500
3	phi_taq	0.715931	500
4	phi_bam	0.823658	0
4	mer_bam	0	0
4	mcs_bam	0	0
4	uwec_bam	1	0
4	phi_bmq	0.752429	0
4	mer_bmq	3	0
4	mcs_bmq	0.833333	0
4	uwec_bmq	0.666667	0
4	phi_boz	0.916823	0
4	mer_boz	0	0
4	mcs_boz	0	0
4	uwec_boz	1	0
4	phi_dtm	0.809387	0
4	mer_dtm	0.5	0
4	mcs_dtm	0.333333	0
4	uwec_dtm	0.666667	0
4	phi_ful	0.980054	0
4	mer_ful	3	0
4	mcs_ful	0.75	0
4	uwec_ful	0.625	0
4	phi_mey	0.768726	0
4	mer_mey	0	0
4	mcs_mey	0	0
4	uwec_mey	0.5	0
4	phi_mku	0.908601	0
4	mer_mku	12	0
4	mcs_mku	0.923077	0
4	uwec_mku	0.538462	0
4	phi_kao	0.970238	0
4	mer_kao	25.3333	0

Run	Metric	Final value	Step
4	mcs_kao	0.974359	0
4	uwec_kao	1	0
4	phi_myk	0.7953	0
4	mer_myk	10	0
4	mcs_myk	0.909091	0
4	uwec_myk	0.5	0
4	phi_ses	0.836952	0
4	mer_ses	0	0
4	mcs_ses	0	0
4	uwec_ses	1	0
4	phi_spp	0.853018	0
4	mer_spp	6.66667	0
4	mcs_spp	0.909091	0
4	uwec_spp	0.545455	0
4	phi_snk	0.781673	0
4	mer_snk	0.333333	0
4	mcs_snk	0.333333	0
4	uwec_snk	0.666667	0
4	phi_taq	0.84844	0
4	mer_taq	0.5	0
4	mcs_taq	0.5	0
4	uwec_taq	1	0
4	phi	0.849638	0
4	mer	5.16667	0
4	mcs	0.865169	0
4	uwec	0.769663	0

Run	Inline claim	Value	Phase
4	phi	0.849638	eval
4	mer	5.16667	eval
4	mcs	0.865169	eval
4	uwec	0.769663	eval

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	2189.8 s
r2	Operating system	linux / amd64
r2	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r2	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r2	Memory	31.0 GB
r2	Total run time	4033.2 s
r3	Operating system	linux / amd64
r3	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r3	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r3	Memory	31.0 GB

Report	Item	Value
r3	Total run time	3993.8 s
r4	Operating system	linux / amd64
r4	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r4	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r4	Memory	31.0 GB
r4	Total run time	44555.7 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	4a6ef8c7ed1f6bcd3c1af4057fd1870532ebb5d4c95cbc7b7fec9701f1458b97	8.8 MB
r1.kvbundle.zip	199eb8b7c51638eae96fa0c03f3f5834f1b16c35debd164a5a4999b69388553a	345 KB
r2.pdf	d6cd2489aa7e223b79fe5ede12864e2a689c1da03093eb7bcf2ed0a25bc35454	9.9 MB
r2.kvbundle.zip	b88486ea00ad32dd4baaed944b0a415e59f7166d1664ca1c60cbe4f3263f521c	346 KB
r3.pdf	442c651ab9985d1b4821bcdbaa2bca5a57c6bf0b7ce707ef98cacf1649a4c7a0	9.8 MB
r3.kvbundle.zip	62f040330bb576322043e72745134b995ff5b696bdfa1f8c62fa006027af6947	348 KB
r4.pdf	f6b8bfd740ffdeb0bdc528f8a74a38e4676d055912825cee7af5059bde02e87d	11.2 MB
r4.kvbundle.zip	01e888a9b7fe0b7054b13f5346d274a03c72df433c3ab9d540027f2be470325	345 KB

5 Links and licence

Code: <https://github.com/mlsftwrs/sebeni>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: nlp, low-resource-languages, manding, morphology, post-training.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.
- [2] Sealed experiment report r1 (SFT arm), kv:2610.00001v2/r1. Data hash 1c1699a5bb61643823134d4b20b8cbf76989b4df4327e1bd9ed91816f2e2d177. <https://kveritas.org/records/2610.00001v2/r1.pdf>
- [3] Sealed experiment report r2 (DPO arm), kv:2610.00001v2/r2. Data hash b2c776c815223b6b9677d4aa01595ece5ca4c79c02fde0bbdd361ba783e7b4d8. <https://kveritas.org/records/2610.00001v2/r2.pdf>
- [4] Sealed experiment report r3 (APO arm), kv:2610.00001v2/r3. Data hash 3d35b38ed6fb3a13c0ac03d020b070894bb4f1d5469f322ef456aca5b4eff9a6. <https://kveritas.org/records/2610.00001v2/r3.pdf>
- [5] Sealed experiment report r4 (GRPO arm), kv:2610.00001v2/r4. Data hash 5eb44231f05c570e8de7328f5ef563ac55d43fb33de1f5a9f1569e3edf0bb93f. <https://kveritas.org/records/2610.00001v2/r4.pdf>